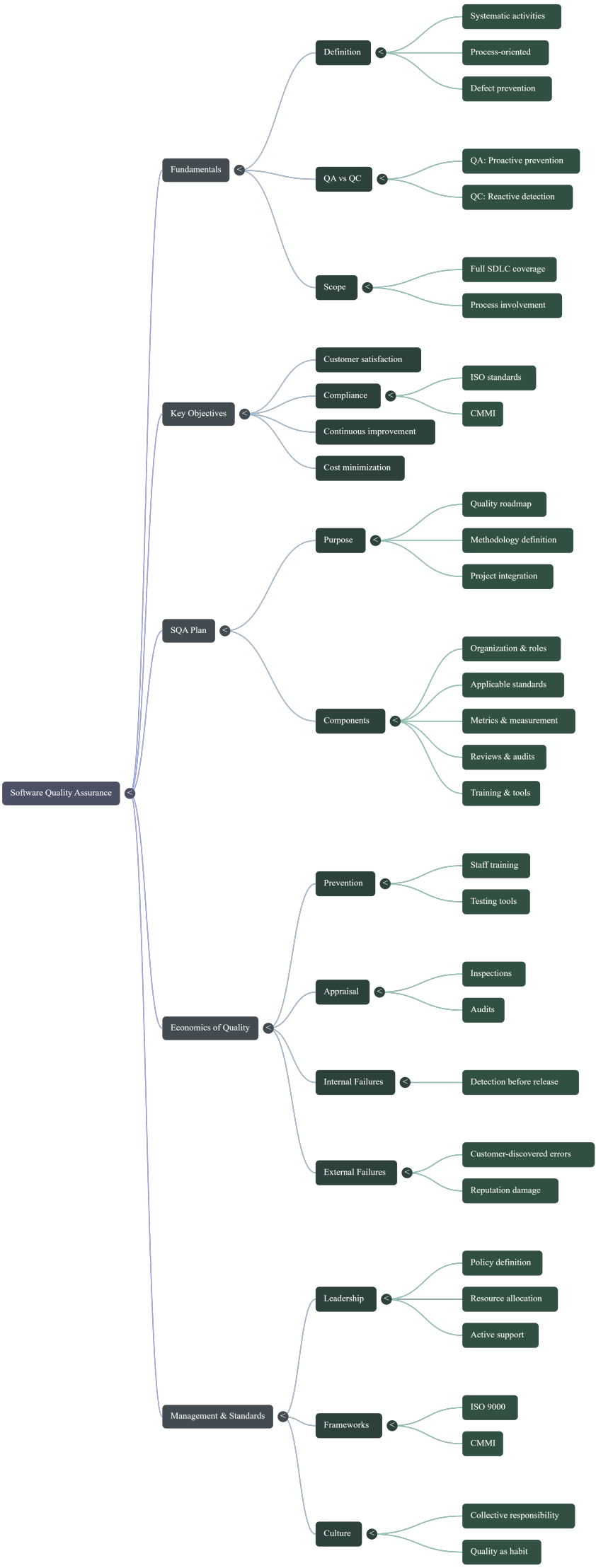

Software Quality Assurance
Fundamentals
Definition
Systematic activities
Process-oriented
Defect prevention
QA vs QC
QA: Proactive prevention
QC: Reactive detection
Scope
Full SDLC coverage
Process involvement
Key Objectives
Customer satisfaction
Compliance
ISO standards
CMMI
Continuous improvement
Cost minimization
SQA Plan
Purpose
Quality roadmap
Methodology definition
Project integration
Components
Organization & roles
Applicable standards
Metrics & measurement
Reviews & audits
Training & tools
Economics of Quality
Prevention
Staff training
Testing tools
Appraisal
Inspections
Audits
Internal Failures
Detection before release
External Failures
Customer-discovered errors
Reputation damage
Management & Standards
Leadership
Policy definition
Resource allocation
Active support
Frameworks
ISO 9000
CMMI
Culture
Collective responsibility
Quality as habit